Muhammad Ibrahim

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Education

New York University — Abu Dhabi, UAE
Bachelor of Science in General Engineering (2:1) — May 2026

- **Modules:** Instrumentation, Sensors and Actuators, Computer-Aided Design, Thermodynamics, Heat Transport
- **Awards:** Full Scholarship, Hyundai NGV Industry Recognition and Grant, Mubadala Research Grant
- **Certifications:** Certified SolidWorks Associate (CSWA), McKinsey Forward Program

NYU Study Away Term — Valletta, Malta
Architecture and Climate Impact (1st Class Honours) — Jun 2023

- Examined how architectural decisions shape climate outcomes, including a site visit to Gozo's Neolithic temples.

Experience

Data Analytics Intern — May 2026 – Jul 2026
Center for Stability, Instability and Turbulence, NYU — *New York, USA*

- Characterised free-surface deformation over a submerged aerofoil through 60 independently planned 2D water-channel trials across four angles of attack.
- Extended predictive range from 15° to 45° and cut estimated project time by 40% using dimensional analysis, Python, MATLAB and machine learning, **avoiding 50+ tunnel runs and saving 40+ hours per week**.

Research Assistant — Apr 2023 – May 2026
Center for Stability, Instability and Turbulence, NYU — *Abu Dhabi, UAE*

- Achieved boundary-layer separation in **100% of water-tunnel runs**, validating five SolidWorks-designed aerofoil and cylinder models and establishing repeatable conditions for experimental testing.
- Identified flow rate and cylinder diameter as key drivers of downstream wake instability through dye-injection flow visualisation.
- Recreated two classical fluid instabilities across 130+ experimental cases with a research partner, contributing data to a co-authored paper now under peer review at the *Journal of Fluid Mechanics*.

Manufacturing and Prototyping Technician — Sep 2025 – May 2026
Advanced Manufacturing Workshop, NYU — *Abu Dhabi, UAE*

- Cut FDM print failures **from 50% to 10%** over 70+ print hours using fabric enclosures and filament-specific settings.
- Delivered eight load-bearing components on schedule for three capstone teams, including a bio-inspired robotic arm, using FEA and calliper inspections to reject 15 defective prints before assembly; composite reinforcement increased printed-part strength by 50%.

Selected Engineering Projects

Compressor-Less Thermoelectric Refrigerator — Jan 2025 – Apr 2025

- Co-developed and assembled an electronically controlled thermoelectric refrigerator targeting 2–12 °C storage, integrating Peltier modules, SolidWorks-designed mounts and thermal-management hardware.
- Achieved cooling **from 27 °C to 10 °C in 24 minutes** and maintained the setpoint within ±0.5 °C sensor accuracy through Arduino-based temperature control; thermal-camera testing guided heat-management refinements.

Advanced Anti-Roll Tank System (ART3S) — Oct 2024 – Sep 2025

- Led mechanical design and system integration for a lower-maintenance ship stabilisation system, modelling 200–800 m^3 anti-roll tanks in SolidWorks and coordinating sensor and hydraulic-actuator interfaces.
- Contributed to MATLAB/Simulink simulations and 29 cm scale-hull wave-tank testing; simulations predicted 85% passive and **over 98%** active reductions in vessel roll amplitude.
- Won a competitive Mubadala research grant for the team by presenting the design to the Mubadala committee.

Traffic-Powered Savonius Wind Turbine — Sep 2024 – May 2025

- Optimised serrated, 45° helical Savonius blades for traffic-wake energy harvesting; ANSYS Fluent CFD predicted **72% higher peak torque and 56% greater power** than a plain-blade baseline.
- Demonstrated **self-starting within 2–5 seconds** in roadside trials, generating a steady 2.7 V baseline and peaks of 10 V using a team-built 1:3 FDM prototype and generator rig.

Skills

Software: SolidWorks, ANSYS Fluent (CFD), MATLAB, Simulink, Python, Arduino
Design: Mechanical design, design for manufacture, prototyping, dimensional analysis, data post-processing
Hands-on: Machine-shop fabrication, 3D printing, composites, sensors, hydraulic actuation, control systems
Testing: Hardware testing, water-tunnel and wave-tank testing, thermal-camera testing, calliper inspection
Affiliations: American Society of Mechanical Engineers (ASME), Student Member
Interests: Varsity jiu-jitsu, wall climbing, golf, padel, mountaineering, biking